BLAST tutorial

Here, there is an additional BLAST tutorial as part of this practical. In this tutorial, we show you how to use a single hypothetical gene sequence (resulting from an ORF prediction) to run a BLAST search and predict what the function of this gene could be by looking at homologues retrieved with BLAST from a selection of other organisms.

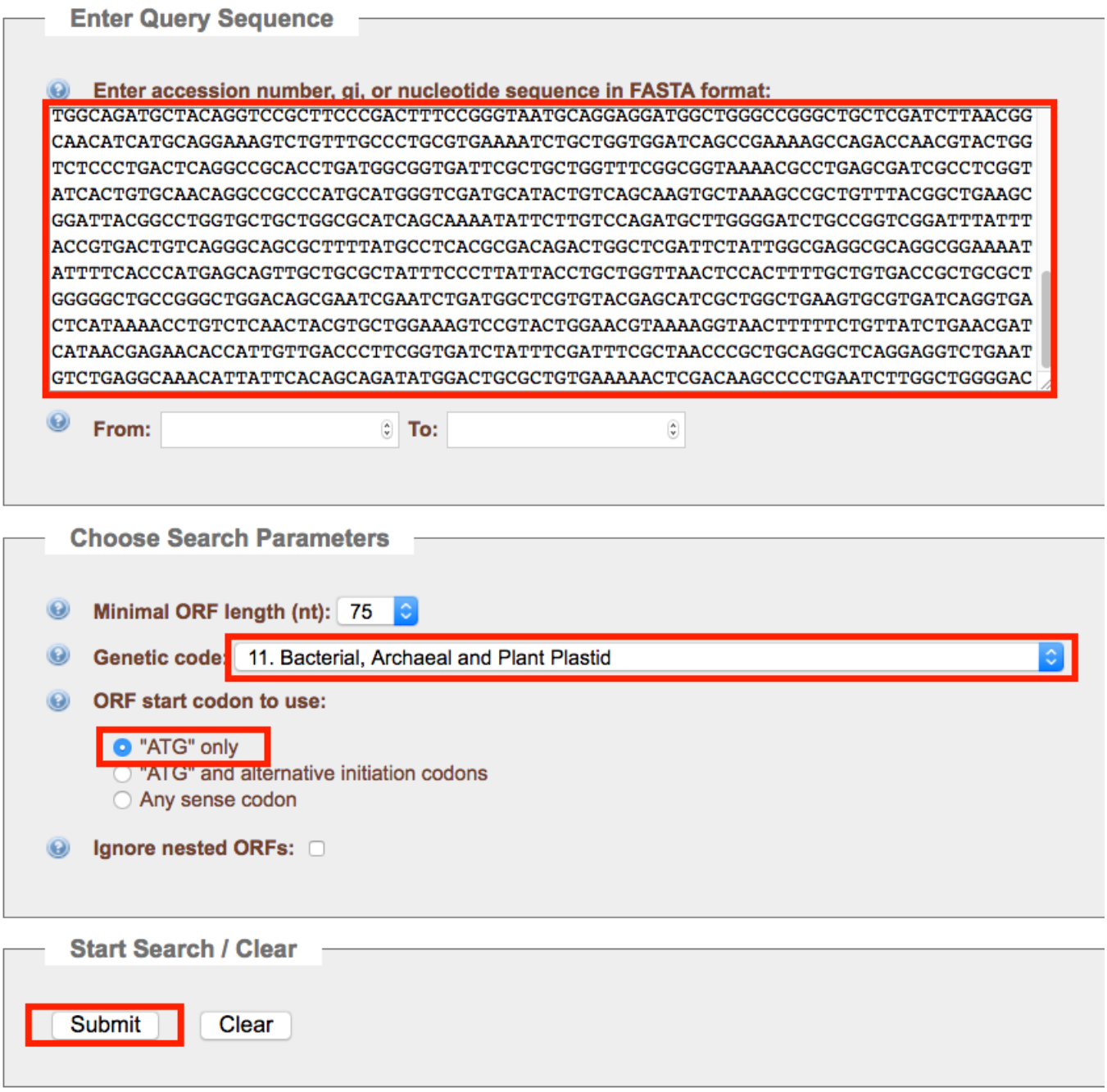
ORF Open Reading Frame prediction

Go to https://www.ncbi.nlm.nih.gov/orffinder/ and introduce below fasta sequence. You can also pick random sequence from "AFPN02.1_merge_consensus.fa" and try to predict open reading frames.

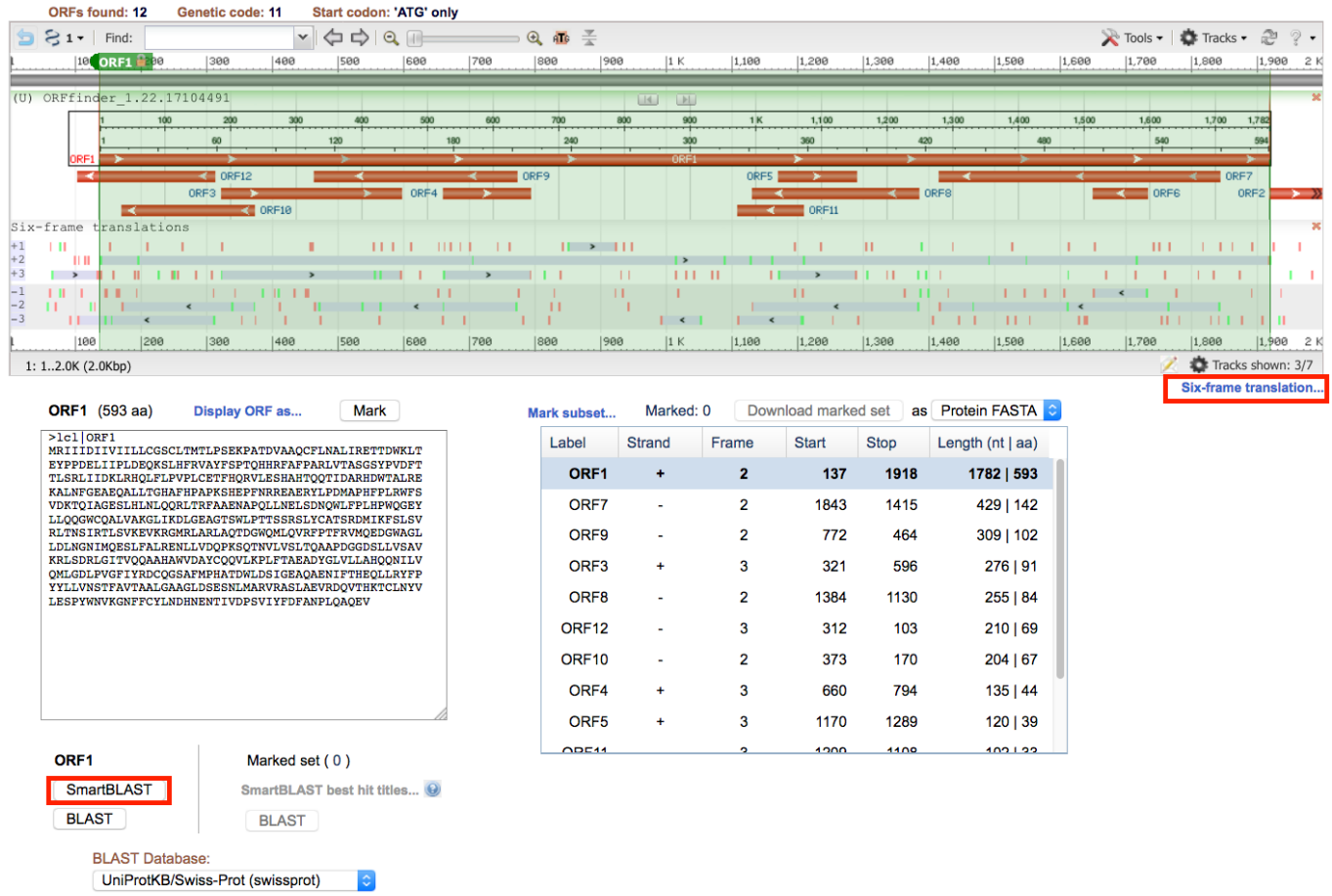
```
>Extracted fasta
ACGCAAGTGGCCAGCAGGACAGTGGTTTTTGGGGCGTATCTTCGATCCCAGGGGACATTTTAATGTTTCAACTC
CATGTA
TTAATTGTGTTTATTTGTAAAATTAATTTATCTGACAATAACATTTCTTATTGATAATGAGAATCATTATTGAC
ATAATT
GTTATTATTTTACTGTGTGGGAGCTGTTTGACTATGACCCTGCCCTCTGAAAAACCAGCCACAGATGTGGCTGC
GCAGTG
CTTCCTGAATGCACTGATTCGTGAAACCACAGACTGGAAACTGACAGAATACCCGCCAGACGAATTGATCATCC
CGCTGG
ATGAGCAGAAATCGCTCCATTTCAGAGTGGCTTATTTCTCCCCAACCCAGCATCACCGCTTTGCATTTCCTGCA
CGTCTG
GTCACGGCATCAGGCAGTTATCCTGTCGACTTTACCACTCTCTCCCGGCTTATTATTGATAAGCTACGGCATCA
ACTGTT
TCTGCCCGTTCCCCTCTGCGAAACTTTCCACCAGCGCGTGCTGGAAAGCCACGCCCATACGCAACAGACAATTG
ATGCCC
GTCATGACTGGACCGCCCTGCGTGAAAAAGCGTTGAATTTTGGCGAAGCTGAGCAGGCGCTGCTGACAGGACAC
GCTTTC
CACCCTGCGCCTAAGTCTCATGAACCGTTTAACCGGCGGGAGGCTGAACGCTACCTGCCTGATATGGCACCCCA
CTTCCC
ACTGCGGTGGTTTTCGGTGGATAAAACGCAAATCGCCGGTGAAAGTTTGCATCTCAACCTTCAACAGCGGCTGA
CGCGAT
TTGCCGCAGAGAATGCGCCTCAGTTACTCAACGAATTAAGTGACAACCAATGGCTGTTCCCGTTGCACCCGTGG
CAGGGG
GAATATCTTTTGCAGCAGGGGTGGTGCCAGGCACTTGTTGCTAAAGGGCTGATTAAAGACTTAGGTGAGGCCGG
CACGTC
GTGGCTGCCGACCACCTCTTCCCGTTCCCTCTACTGTGCCACCAGCCGCGATATGATCAAGTTCTCCCTGAGCG
TACGGC
TGACCAACTCCATCCGTACTCTGTCCGTGAAAGAAGTGAAGCGTGGGATGCGCCTGGCACGCCTGGCTCAAACC
GACGGC
TGGCAGATGCTACAGGTCCGCTTCCCGACTTTCCGGGTAATGCAGGAGGATGGCTGGGCCGGGCTGCTCGATCT
TAACGG
CAACATCATGCAGGAAAGTCTGTTTGCCCTGCGTGAAAATCTGCTGGTGGATCAGCCGAAAAGCCAGACCAACG
TACTGG
TCTCCCTGACTCAGGCCGCACCTGATGGCGGTGATTCGCTGCTGGTTTCGGCGGTAAAACGCCTGAGCGATCGC
CTCGGT
ATCACTGTGCAACAGGCCGCCCATGCATGGGTCGATGCATACTGTCAGCAAGTGCTAAAGCCGCTGTTTACGGC
```

```
TGAAGC
GGATTACGGCCTGGTGCTGCTGGCGCATCAGCAAAATATTCTTGTCCAGATGCTTGGGGATCTGCCGGTCGGAT
TTATTT
ACCGTGACTGTCAGGGCAGCGCTTTTATGCCTCACGCGACAGACTGGCTCGATTCTATTGGCGAGGCGCAGGCG
GAAAAT
ATTTTCACCCATGAGCAGTTGCTGCGCTATTTCCCTTATTACCTGCTGGTTAACTCCACTTTTGCTGTGACCGC
TGCGCT
GGGGGCTGCCGGGCTGGACAGCGAATCGAATCTGATGGCTCGTGTACGAGCATCGCTGGCTGAAGTGCGTGATC
AGGTGA
CTCATAAAACCTGTCTCAACTACGTGCTGGAAAGTCCGTACTGGAACGTAAAAGGTAACTTTTTCTGTTATCTG
AACGAT
CATAACGAGAACACCATTGTTGACCCTTCGGTGATCTATTTCGATTTCGCTAACCCGCTGCAGGCTCAGGAGGT
CTGAAT
GTCTGAGGCAAACATTATTCACAGCAGATATGGACTGCGCTGTGAAAAACTCGACAAGCCCCTGAATCTTGGCT
GGGGAC
```

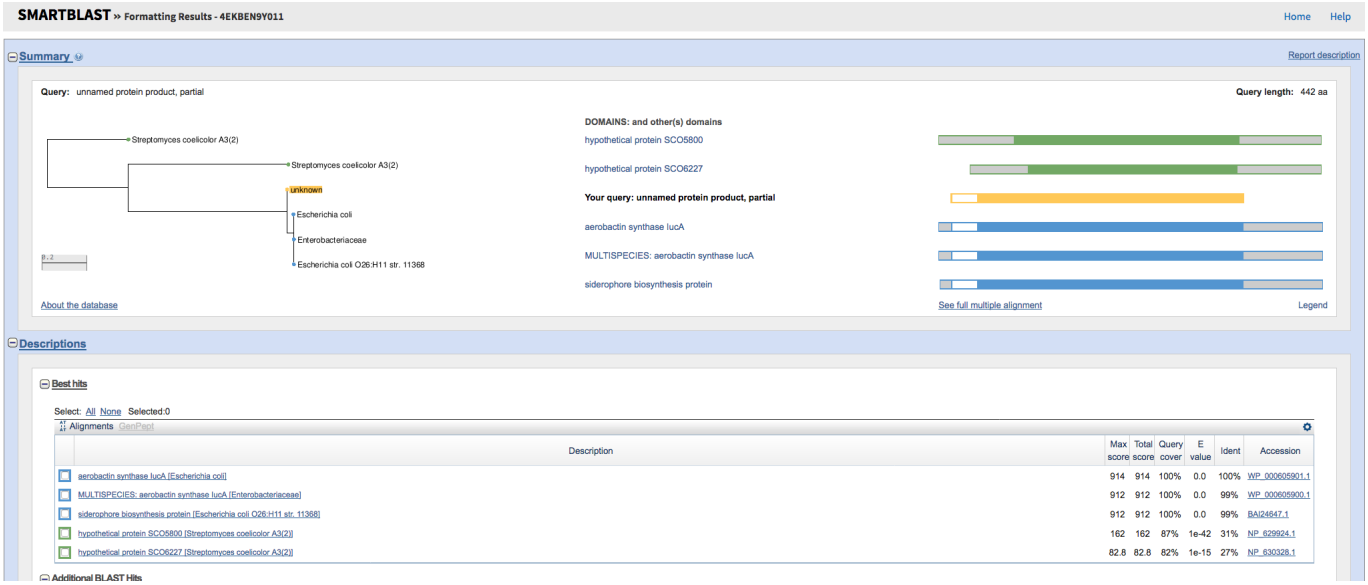
Select bacterial genetic code. For ORF start codon to use, select "ATG" only and submit.



Select the longest ORF, show the six frame translation and SmartBlast.



Explore the BLAST phylogenetic tree and aligned sequence.



![BLAST output](figures/blast _output.png)

BLAST algorithm

https://en.wikipedia.org/wiki/BLAST

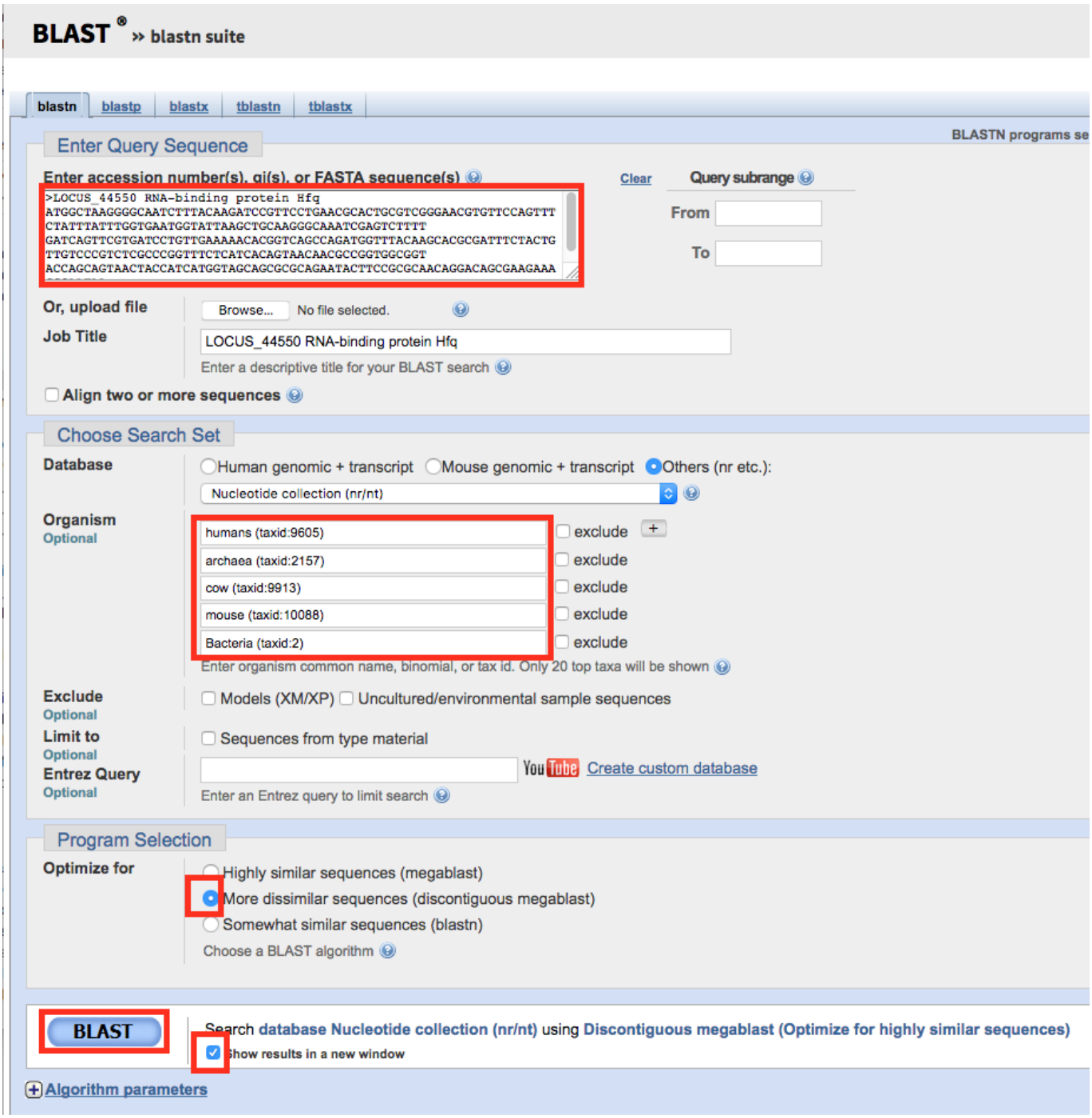
Used to seach DNA or Protein sequences on biological databases (NCBI, ENSEMBL...). You can also create your own database to BLAST localy.

BLAST nucleotide sequence

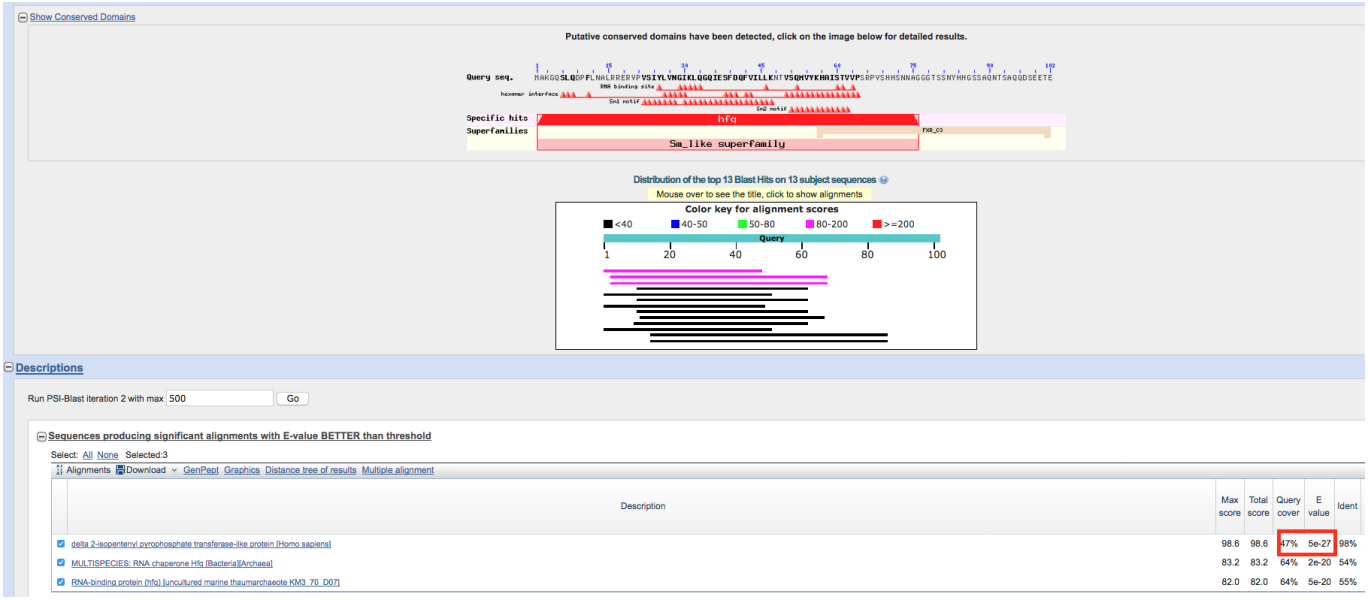
Go to https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE_TYPE=BlastSearch

Insert Hfq fasta sequence - extracted from our consensus sequence - and select organism to perform BLAST (humans, archaea, cow, mouse, bacteria). Select "discontiguous megablast" (selected for evolutive distant species) and "show results in a new window".

```
>LOCUS_44550 RNA-binding protein Hfq
ATGGCTAAGGGGCAATCTTTACAAGATCCGTTCCTGAACGCACTGCGTCGGGAACGTGTTCCAGTTTCTATTTA
TTTGGTGAATGGTATTAAGCTGCAAGGGCAAATCGAGTCTTTT
GATCAGTTCGTGATCCTGTTGAAAAACACGGTCAGCCAGATGGTTTACAAGCACGCGATTTCTACTGTTGTCCC
GTCTCGCCCGGTTTCTCATCACAGTAACAACGCCGGTGGCGGT
ACCAGCAGTAACTACCATCATGGTAGCAGCGCGCAGAATACTTCCGCGCAACAGGACAGCGAAGAAACCGAATA
A
```



Click on human alignment. Surprised on the level of conservation between Eukaryotes and bacteria!!!! 😮

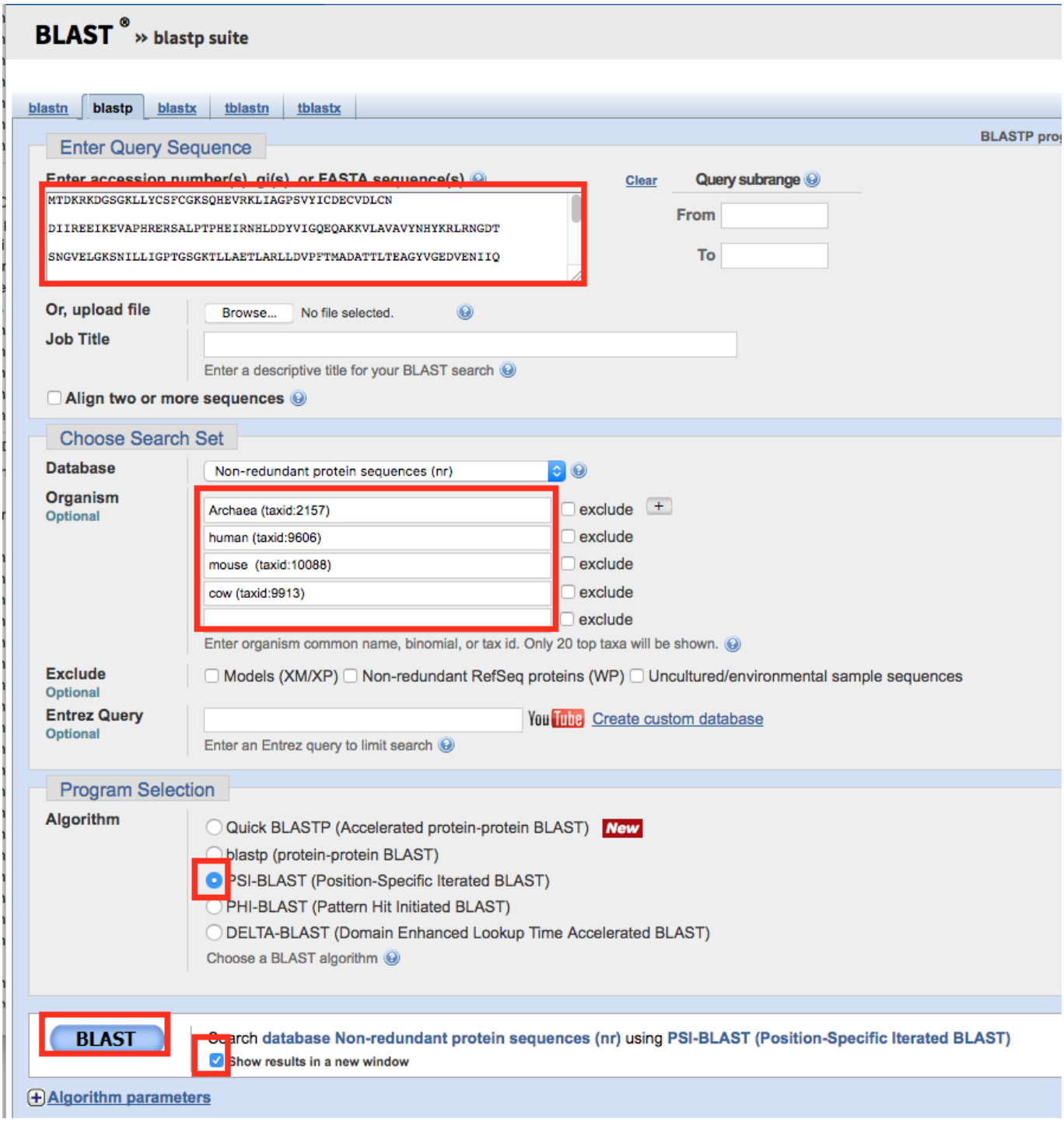


BLAST protein sequence

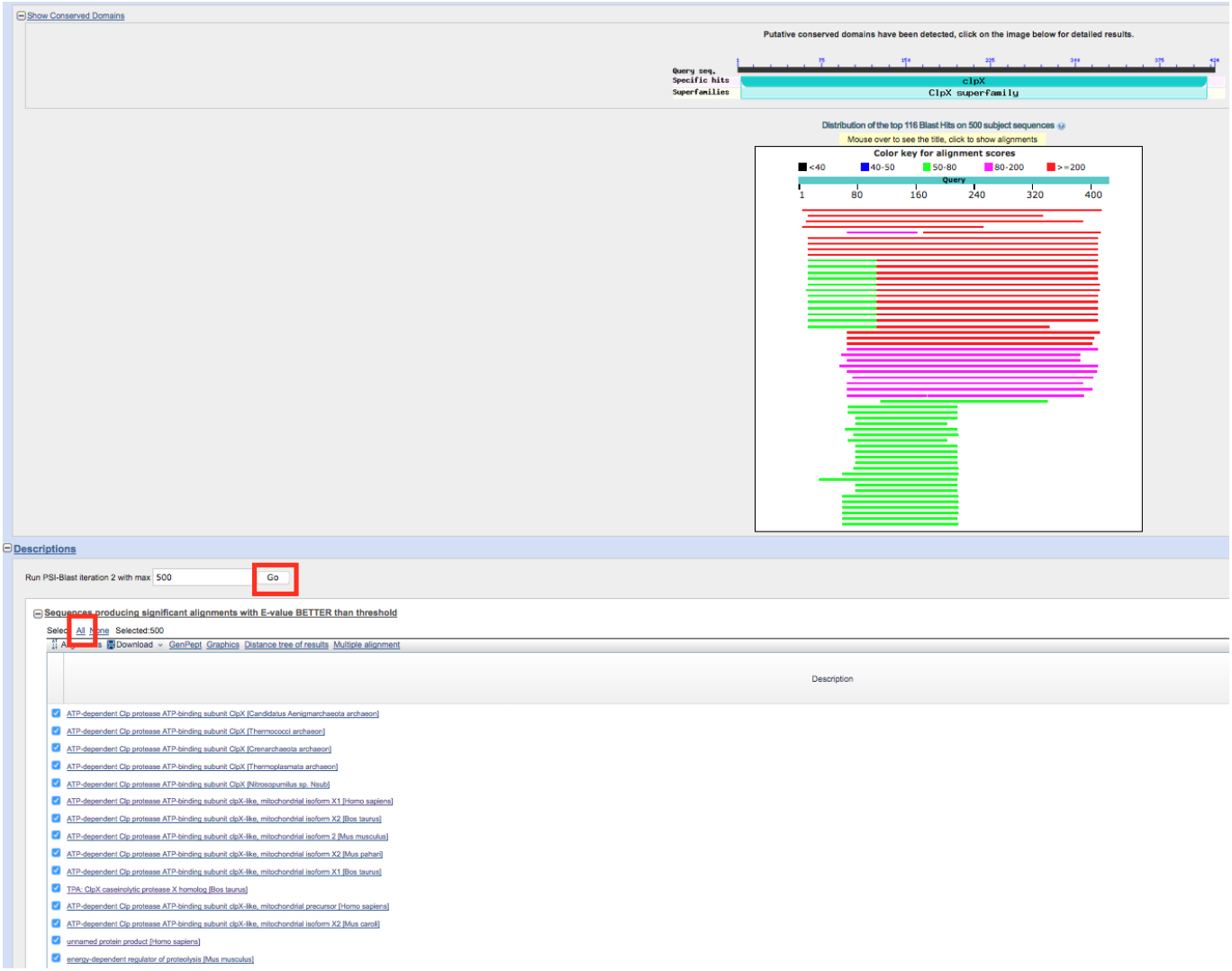
Go to https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Proteins

Insert ClpX protein fasta sequence - extracted from our annotated sequence - and select organism to perform BLAST (humans, archaea, cow, mouse). Select "PSI-BLAST" (Position-Specific Iterated BLAST) and "show results in a new window".

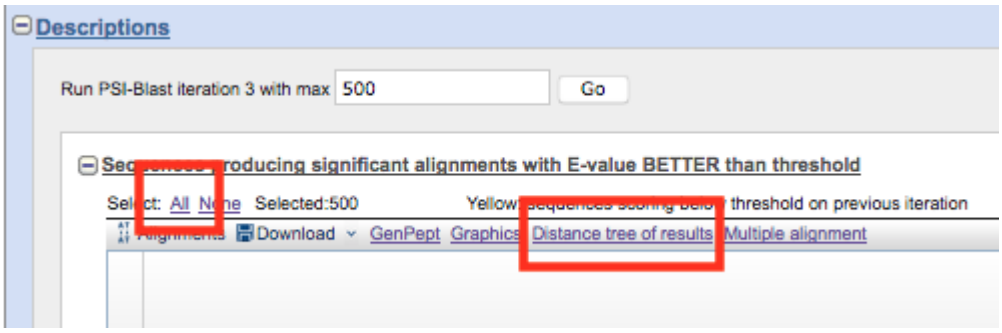
```
> ATP-dependent protease ATP-binding subunit ClpX from AFPN02.1
MTDKRKDGSGKLLYCSFCGKSQHEVRKLIAGPSVYICDECVDLCN
DIIREEIKEVAPHRERSALPTPHEIRNHLDDYVIGQEQAKKVLAVAVYNHYKRLRNGDT
SNGVELGKSNILLIGPTGSGKTLLAETLARLLDVPFTMADATTLTEAGYVGEDVENIIQ
KLLQKCDYDVQKAQRGIVYIDEIDKISRKSDNPSITRDVSGEGVQQALLKLIEGTVAAV
PPQGGRKHPQQEFLQVDTSKILFICGGAFAGLDKVISHRVETGSGIGFGATVKAKSDKA
SEGELLAQVEPEDLIKFGLIPEFIGRLPVVATLNELSEEALIQILKEPKNALTKQYQAL
FNLEGVDLEFRDEALDAIAKKAMARKTGARGLRSIVEAALLDTMYDLPSMEDVEKVVID
ESVIDGQSKPLLIYGKPEAQQASGE
```



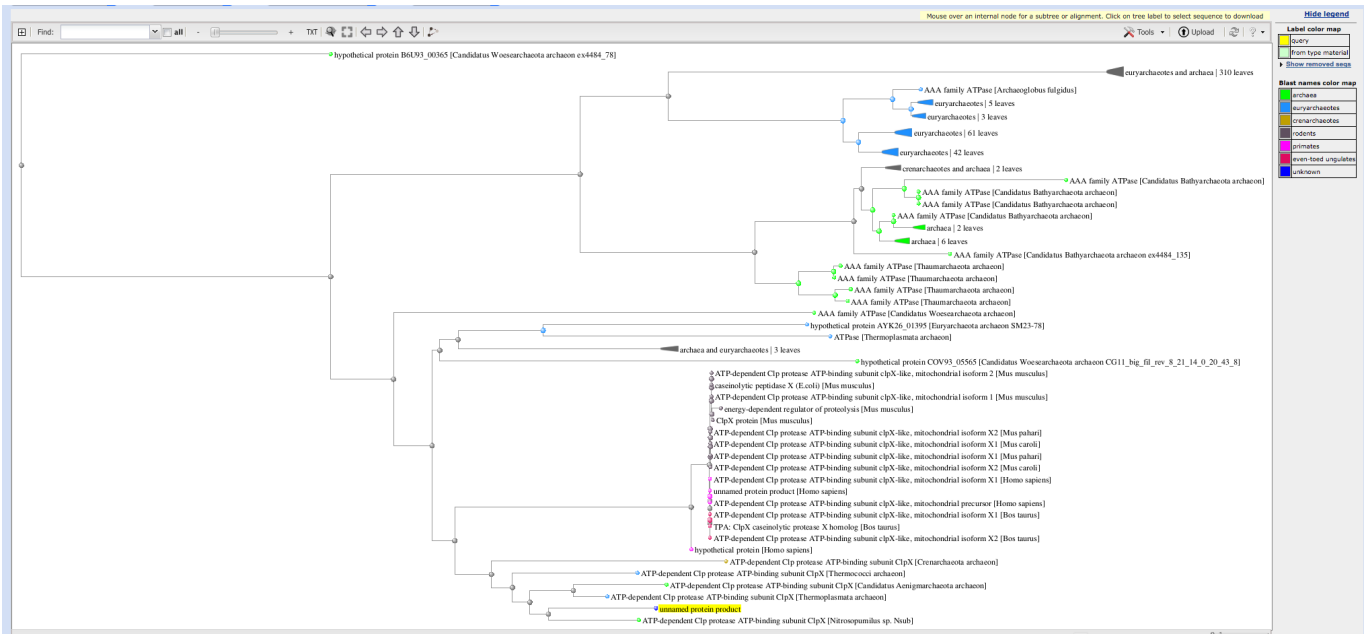
Select "All" and run 2nd PSI_BLAST Iteration.



Select "All" and "Distance tree of results".



Observe pylogenetic tree using only one gene!!



Homologous paralogous

https://www.ncbi.nlm.nih.gov/homologene

http://pfam.xfam.org/family/PF00816